www.Padasalai.Net www.TrbTnpsc.com

V12M

Virudhunagar District

Common Quarterly Examination - September 2026

Standard 12

Time Allowed: 3.00 Hours **MATHEMATICS** Maximum Marks: 90

Part - I

I. Choose the correct answer: 20 x 1 = 20

1. If $A=\begin{bmatrix}3 & 5\\ 1 & 2\end{bmatrix}$, B = adj A and C = 3A, then $\frac{|adj\ B|}{|C|}$ =

   a) $\frac{1}{3}$ b) $\frac{1}{9}$ c) $\frac{1}{4}$ d) 1

2. If A^TA^{-1} is symmetric, then A^2 =

   a) A^{-1} b) $(A^T)^2$ c) A^T d) $(A^{-1})^2$

3. If $A=\begin{bmatrix}\cos\theta & \sin\theta\\ -\sin\theta & \cos\theta\end{bmatrix}$ and $A(adj\ A)=\begin{bmatrix}k & 0\\ 0 & k\end{bmatrix}$, then k =

   a) 0 b) $\sin\theta$ c) $\cos\theta$ d) 1

4. If p represents the rank and, A and B are n x n matrices, then

   a) $\rho(A+B) = \rho(A) + \rho(B)$ b) $\rho(AB) = \rho(A)\ \rho(B)$

   c) $\rho(A-B) = \rho(A) - \rho(B)$ d) $\rho(A+B) \le n$

5. The solution of the equation $|z| - z = 1 + 2i$ is

   a) $\frac{3}{2}-2i$ b) $-\frac{3}{2}+2i$ c) $2-\frac{3}{2}i$ d) $2+\frac{3}{2}i$

6. The value of $\left(\frac{1+\sqrt{3}i}{1-\sqrt{3}i}\right)^{10}$ is

   a) $cis\frac{2\pi}{3}$ b) $cis\frac{4\pi}{3}$ c) $-cis\frac{2\pi}{3}$ d) $-cis\frac{4\pi}{3}$

7. If $\omega \neq 1$ is a cubic root of unity and $(1+\omega)^7 = A+B\omega$, then(A, B) equals

   a) (1,0) b) (-1,1) c) (0,1) d) (1,1)

8. If f and g are polynomials of degree is m and n respectively, and if h(x) = (fog) (x), then the degree of h is

   a) mn b) m+n c) m^n d) n^m

9. According to the rational root theorem, which number is not a possible rational zero of $4x^7 + 2x^4 - 10x^3 - 5$ = ?

   a) -1 b) $\frac{5}{4}$ c) $\frac{4}{5}$ d) 5

10. If $\cot^{-1}2$ and $\cot^{-1}3$ are two angles of a triangle, then the third angle is

    a) $\frac{\pi}{4}$ b) $\frac{3\pi}{4}$ c) $\frac{\pi}{6}$ d) $\frac{\pi}{3}$

11. If $\sin^{-1}\frac{x}{5}+\text{cosec}^{-1}\frac{5}{4}=\frac{\pi}{2}$, then the value of x is

    a) 4 b) 5 c) 2 d) 3

Padasalai.Net

Scanned with OKEN Scanner

Kindly Send Me Your Study Materials To Us Email ID: padasalai.net@gmail.com

www.Padasalai.Net www.TrbTnpsc.com

12. If $\cot^{-1}x=\frac{2\pi}{5}$, for some $x\in R$, the value of $\tan^{-1}x$ is

a) $\frac{-\pi}{10}$ b) $\frac{-\pi}{5}$ c) $\frac{\pi}{10}$ d) $\frac{\pi}{5}$

13. If P(x, y) be any point on $16x^2+25y^2=400$ with foci $F_1(3,0)$ and $F_2(-3,0)$, then $PF_1 + PF_2$ is

a) 8 b) 6 c) 10 d) 12

14. The values of m for which the line $y=mx+2\sqrt{5}$ touches the hyperbola $16x^2-9y^2=144$ are the roots of $x^2-(a+b)x-4=0$, then the value of (a+b) is

a) 2 b) 4 c) 0 d) -2

15. The eccentricity of the hyperbola whose latus rectum is 8 and conjugate axis equal to half the distance between the foci is

a) $4/3$ b) $4/\sqrt{3}$ c) $2/\sqrt{3}$ d) $3/2$

16. If $y=mx+c$ is a tangent to the parabola $y^2=4ax$, then the point of contact is

a) $\left(\frac{a}{m^2},\frac{2a}{m}\right)$ b) $\left(\frac{-a}{m^2},\frac{2a}{m}\right)$ c) $\left(\frac{a}{m^2},\frac{-2a}{m}\right)$ d) $\left(\frac{-a}{m^2},\frac{-2a}{m}\right)$

17. If $\vec{a}$ and $\vec{b}$ are parallel vectors, then $[\vec{a}\ \vec{c}\ \vec{b}]$ is equal to

a) 2 b) -1 c) 1 d) 0

18. If the direction cosines of a line are $\frac{1}{c},\frac{1}{c},\frac{1}{c}$, then

a) $c=\pm3$ b) $c=\pm\sqrt{3}$ c) $c>0$ d) $0<c<1$

19. If the length of the perpendicular from the origin to the plane $2x+3y+\lambda z=1$, $\lambda>0$ is $1/5$, then the value of λ is

a) $2\sqrt{3}$ b) $3\sqrt{2}$ c) 0 d) 1

20. The non-parametric vector equation of the plane passing through the three points whose position vectors are $\vec{a},\vec{b}$ and $\vec{c}$ is / are

i) $[\vec{r}-\vec{a}\ \ \vec{b}-\vec{a}\ \ \vec{c}-\vec{a}]=0$ ii) $[\vec{r}-\vec{a}\ \ \vec{a}-\vec{b}\ \ \vec{c}-\vec{a}]=0$

iii) $[\vec{r}-\vec{a}\ \ \vec{b}-\vec{a}\ \ \vec{a}-\vec{c}]=0$ iv) $[\vec{r}-\vec{a}\ \ \vec{a}-\vec{b}\ \ \vec{a}-\vec{c}]=0$

a) (ii) and (iii) b) (iii) and (iv) c) all d) none

Part - II

II. Answer any 7 questions. (Q.No.30 is compulsory) **7 x 2 = 14**

21. If Z = (2+3i) (1-i), find z^{-1}

22. Form a polynomial equation with integer coefficients with $\sqrt{\frac{\sqrt{2}}{\sqrt{3}}}$ as a root.

23. If α and β are the roots of the equation $x^2-5x+6=0$, then prove that $\alpha^2-\beta^2=\pm5$

24. Find the modulus and principal argument of the complex number $3-i\sqrt{3}$

25. State The reason for $\cos^{-1}\left[\cos\left(\frac{-\pi}{6}\right)\right]\neq\frac{-\pi}{6}$

Padasalai.Net

Scanned with OKEN Scanner

Kindly Send Me Your Study Materials To Us Email ID: padasalai.net@gmail.com

www.Padasalai.Net www.TrbTnpsc.com

26. Find the rank of the matrix $\begin{bmatrix} 1 & -2 & -1 & 0 \\ 3 & -6 & -3 & 1 \end{bmatrix}$

27. Find the general equation of a circle with centre (-3,-4) and radius 3 units.

28. Find the equation of the tangent to the parabola $y^2=8x$ at $t=2$

29. Show that the vectors $\hat{i}+2\hat{j}-3\hat{k}$, $2\hat{i}-\hat{j}+2\hat{k}$ and $3\hat{i}+\hat{j}-\hat{k}$ are coplanar.

30. $\begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$ is orthogonal. - Prove it.

Part - III

III. Answer any 7 questions. (Q.No.40 is compulsory) 7 x 3 = 21

31. If any ΔABC, prove by vector method that $\frac{a}{\sin A}=\frac{b}{\sin B}=\frac{c}{\sin C}$

32. Find the angle between the line $\vec{r}=(2\hat{i}-\hat{j}+\hat{k})+t(\hat{i}+2\hat{j}-2\hat{k})$ and the plane $\vec{r}\cdot(6\hat{i}+3\hat{j}+2\hat{k})=8$

33. Verify $(A^T)^{-1}=(A^{-1})^T$ for the matrix $A=\begin{bmatrix} 2 & 9 \\ 1 & 7 \end{bmatrix}$

34. Solve the system of linear equations $\frac{3}{x}+2y=12$, $\frac{2}{x}+3y=13$ by Cramer's rule.

35. If $\frac{1+z}{1-z}=\cos 2\theta+i\sin 2\theta$, show that $z=i\tan\theta$

36. If two complex numbers, $-3+ix^2y$ and x^2+y+4i are conjugate to each other, then find the values of x and y

37. Solve the equation $x^4-9x^2+20=0$

38. Find the value of $\tan^{-1}(-1)+\cos^{-1}\left(\frac{1}{2}\right)+\sin^{-1}\left(\frac{-1}{2}\right)$

39. Solve: $\tan^{-1}\left(\frac{1-x}{1+x}\right)=\frac{1}{2}\tan^{-1}x$ for $x>0$

40. A circle of area 9π square units has two of its diameters along the lines $x+y=5$ and $x-y=1$. Find the equation of the circle.

Part - IV

IV. Answer all the questions. 7 x 5 = 35

41. a) If $A=\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$, verify that $A(\text{adj } A)=(\text{adj } A)A=|A|I_3$

(OR)

b) Let z_1, z_2 and z_3 be complex numbers such that $|z_1|=|z_2|=|z_3|=r>0$ and $z_1+z_2+z_3\neq 0$. Prove that $\left|\frac{z_1z_2+z_2z_3+z_3z_1}{z_1+z_2+z_3}\right|=r$

Padasalai.Net

Scanned with OKEN Scanner

Kindly Send Me Your Study Materials To Us Email ID: padasalai.net@gmail.com

www.Padasalai.Net www.TrbTnpsc.com

42. a) Draw the graph of $y=\sin x$ in $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ and $y=\sin^{-1}x$ in $[-1,1]$

(OR)

b) Find the equation of the circle passing through the points (1,1), (2,-1) and (3,2)

43. a) Find the value of k for which the equations $kx-2y+z=1$, $x-2ky+z=-2$, $x-2y+kz=1$ have (i) no solution (ii) unique solution (iii) infinitely many solution.

(OR)

b) If $\vec{a}=2\hat{i}+3\hat{j}-\hat{k}$, $\vec{b}=-2\hat{i}+5\hat{k}$, $\vec{c}=\hat{j}-3\hat{k}$, verify that $\vec{a}\times(\vec{b}\times\vec{c})=(\vec{a}.\vec{c})\vec{b}-(\vec{a}.\vec{b})\vec{c}$

44. a) Solve : $(x-4)(x-7)(x-2)(x+1)=16$

(OR)

b) On lighting a rocket cracker it gets projected in a parabolic path and reaches a maximum height of 4m when it is 6 m away from the point of projection. Finally it reaches the ground 12 am away from the starting point. Find the angle of projection.

45. a) Solve the equation $z^3+8i=0$, where $z\in C$

(OR)

b) Prove that $\cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta$ by vector method.

46. a) Solve : $\tan^{-1}\left(\frac{x-1}{x-2}\right)+\tan^{-1}\left(\frac{x+1}{x+2}\right)=\frac{\pi}{4}$

(OR)

b) Identify the type of conic and find the centre, foci, vertices and directrices of the equation $9x^2-y^2-36x-6y+18=0$

47. a) If $2+i$ and $3-\sqrt{2}$ are roots of the equation $x^6-13x^5+62x^4-126x^3+65x^2+127x-140=0$, find all roots.

(OR)

b) Find the non-parametric form of vector equation and the cartesian equation of the plane containing the line $\frac{x-2}{2}=\frac{y-2}{3}=\frac{z-1}{-2}$ and passing through the point (-1,1,-1)

****

Padasalai.Net

Scanned with OKEN Scanner

Kindly Send Me Your Study Materials To Us Email ID: padasalai.net@gmail.com